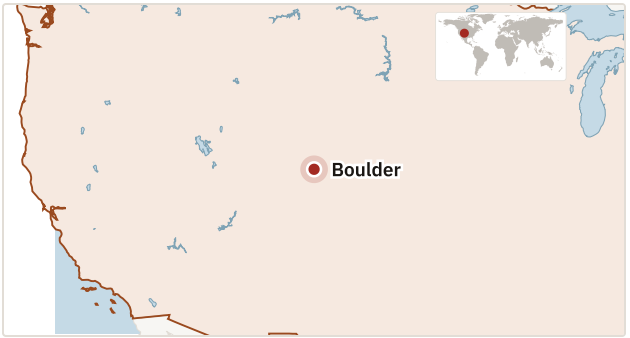
Boulder, Colorado

El Niño's signal in Boulder is heavy, wet snow in late winter. Boulder's killers mostly sit outside it. Downslope windstorms drive grass fire and power shutoffs every winter regardless of El Niño, and drugs kill about twenty-five times as many people as exposure. The most important step is to treat every winter wind day as a fire day, and to put naloxone in the hands of the people around the ones most at risk.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Boulder County, CO (inferred)	None published for the US, Colorado, or the county	n/a here; global 6-month figure 239,000 ± 11,000	Winter wind: grass/WUI fire and power shutoffs. El Niño adds heavy late-winter snow

14/14 recommendations checked. 12 corrected, 1 confirmed as written (multilingual alert sign-up through the San Lazaro steering committee), 1 unverified (parametric and international funds, accepted as not applicable without a source), and 0 contraindicated as actions. Two lens claims were refuted. Alerts are no longer opt-in only, because the county now lists wireless emergency alerts as automatic. And the cold-weather shelter trigger is not too high; the gap is beds. The three funding placements were checked separately: all three corrected, and the naloxone placement redirected from kits to distribution. The funding target fell 22%, from $115k to $90k, inside the one-third flag. **Read the verdicts with care:** the shared web-search budget was exhausted before this run began, so lenses and verifiers fetched primary pages directly; Xcel, PUC, DOI, and several

EXECUTIVE SUMMARY — AT A GLANCE



40.01°N 105.27°W, United States of America · Natural Earth

- **The risk.** Winter wind drives grass fire and power shutoffs regardless of El Niño, and drugs kill about twenty-five times as many people as exposure; El Niño adds heavy late-winter snow.
- **First action.** Boulder County Public Health's The Works Program puts bystander naloxone and training where the city's fentanyl deaths happen, using free state supply, with staff and shelter stock in place by 1 December.*
- **The money.** $50k floor, $90k target, about $200k ceiling; the floor pays $25k for naloxone distribution and $25k for a trigger fund.
- **Do not.** Do not stand down fire readiness because "El Niño means a snowy winter"; treat every winter wind day as a fire day.

What to do	**9** actions	p. 2
Who does what	RACI chart	p. 10
Where money does the most	funding	p. 10
Do not do this	**3** items	p. 12
What has happened here	the record	p. 13
How this was produced	methodology	p. 15
References	sources	p. 16

* Suggested formulation: check with a health worker before sharing.

city pages could not be read. Exposure: Climate Impact Lab, 451,000 at Risk, September 2026, pre-peer-review. It publishes no figure for the United States, Colorado, or Boulder County, and heat is not this place's winter hazard.

Read this first: El Niño is the wrong frame for Boulder, and the plan says so. NOAA's Climate Prediction Center gives a strong-to-very-strong El Niño through January–March 2027 (over 83% chance), and NWS Boulder says El Niño raises the odds of above-average precipitation in northeast Colorado from November to April through upslope storms. The record bears that out in snow: 23.5 inches in December 1982, 30.1 in October 1997, and 34.7 in March 2003, all El Niño months. But the hazards that kill are elsewhere. 74% of Boulder's documented 70-mph-plus wind events fall between November and February, and the Marshall Fire burned on 30 December 2021, in a La Niña winter. In 2025 the county coroner recorded 77 drug and alcohol deaths and 3 environmental-exposure deaths. Every lens that looked at the numbers reached the same view: an El Niño winter here should not lower anyone's guard against wind and fire, and should not draw money away from overdose.

WHAT THIS PLACE SHOULD DO — RANKED

Actions are ranked by expected lives saved per unit of effort before the peak. The backing lenses are shown on each action, and the RACI chart below assigns each one. An action several lenses reached independently is the stronger bet, but only where its evidence came from somewhere like here. One rule underlies every action: the trigger is the wind forecast and the thermometer, never the season's label.

1. Put bystander naloxone and training where the City of Boulder's fentanyl deaths happen, using free state supply

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST · FRONTLINE | INVENTORY: LOOSE ONLY (SUBSTANCE-USE REGULATION)

OWNER	BY WHEN	COST	PREVENTS
Boulder County Public Health's The Works Program (Boulder site at Boulder County AIDS Project, 2118 14th St)	Distribution staff and shelter stock in place by 1 Dec	$25k for distribution and training (ceiling $50k); kits from the state fund	Fentanyl deaths, the county's largest preventable pool

* Suggested formulation: check with a health worker before sharing.

The county coroner recorded 77 drug and alcohol deaths in 2025, with fentanyl the cause in 33. From January to June 2026 there were 37 drug deaths, up 11%. Of the 20 fentanyl deaths, 13 were in the City of Boulder. CDPHE's Naloxone Bulk Purchase Fund has given naloxone free to syringe services programmes, local government, schools, and community organisations since 2019, so the gap is people to hand it out, not kits. Fund distribution and training through The Works, stock All Roads and the critical-weather sites, and pair naloxone with every cold-night outreach round. Since May 2024 the state fund has prioritised supply because money is limited, so confirm The Works' current allocation before buying anything.*

Transferability: the mortality evidence is pre-fentanyl, from Massachusetts (Walley et al., *BMJ* 2013: adjusted death rate ratio 0.54 at high uptake) and a US model (Coffin & Sullivan 2013: 1 death averted per 227 kits); neither was re-checked here. Corrections: The Works is a county programme, and Boulder County AIDS Project is one of its sites. The $232,430 settlement grant to The Naloxone Project is for first-responder distribution, not bystanders. The 2025–26 opioid settlement table totals about $3.95M across 23 lines, not the $4.61M a lens cited.

2. Pre-approve every alert before the wind season: zones, WEA polygons, templates, a live drill, and surge senders

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · HISTORIAN | INVENTORY: CHANGE IN FORECAST QUALITY (SHRADER ET AL. 2023), INDIRECT

OWNER	BY WHEN	COST	PREVENTS
Boulder Office of Disaster Management (ODM) with BRETSA, the regional 911 authority	**15 Nov**	**Staff time inside existing contracts**	**A repeat of the Marshall Fire's 41-minute alert gap**

The Marshall Fire was reported at 11:06 on 30 December 2021. The first mandatory evacuation alert went out at 11:47, to 215 contacts, and cell phones received alerts only if they had been registered. Boulder County had been licensed for wireless emergency alerts (WEA) since 2019, but setup had been shelved by COVID, the King Soopers shooting, a cyberattack, and the Calwood Fire. The director said afterwards, “I wish we would have had WEA implemented.” About 24,289 contacts were eventually alerted. Two people died and 1,084 homes were destroyed. **That gap has since been closed on paper:** boulderodm.gov now lists WEA and EAS as automatic channels for “imminent life threat,” with no opt-in. What remains is speed: pre-approved zones, polygons, and message text for wind-driven WUI fire, Boulder Creek flood, and power shutoffs; a live WEA drill; more than two people at the sending desk; and a confirmed backup sender at the state or a neighbouring county. Keep door-to-door evacuation and Everbridge as layers, because cell service degrades in fires.

* Suggested formulation: check with a health worker before sharing.

Transferability: native to Boulder. Corrections: the gap was about 41 minutes, not 42. The figures “4,637 (19%) confirmed received,” “one dispatcher and one supervisor,” and “the county declined the state's offer to send” came from lenses and could not be found in the sources reached, so they are not used. Whether WEA has actually been *used* in a fire since 2022 was not confirmed. The concern that WEA floods 911 is real and dates from 2022.

3. Treat every winter wind day as a fire day, snow or not

PARTIALLY CONFIRMED | HISTORIAN · SKEPTIC · EMERGENCY MGR | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Boulder Fire-Rescue and ODM, with Xcel for shutoff coordination	**Standing rule from 1 Nov to the end of March**	**Staff time**	**Lowering readiness because “El Niño means snow”**

NOAA's Boulder wind record lists 194 of 262 documented events with gusts of 70 mph or more (74%) between November and February. The highest at NCAR were 147 mph on 23 January 1971 and 137 mph on 16–17 January 1982. The Marshall Fire burned in a snowless late December on grass, with gusts to 115 mph. ODM's own status page records a Highway 287/Dillon Road wildfire with evacuation orders on 5 January 2026, an Xcel power shutoff on 20 December 2025, and a red-flag shutoff on 14 March 2026. El Niño raises the odds of wetter months, but it does not stop wind arriving between snows. The rule: Red Flag and High Wind Warnings trigger fire-weather staffing, pre-positioned crews, and pre-drafted alerts (action 2) all winter, with no seasonal exemption.

* Suggested formulation: check with a health worker before sharing.

Transferability: native. NWS notes its El Niño odds say nothing about the size of the precipitation. The lens report of a Skunk Canyon Fire on 7 October 2026 could not be confirmed, because the ODM page showed nothing after March 2026.

4. Close the medical gap in power shutoffs: enrol the unenrolled, add charging sites, and run buddy lists in the parks and foothills

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · FRONTLINE · ECONOMIST | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Boulder Fire-Rescue and Boulder County Public Health; Xcel Energy; ODM for sites; Boulder County with the San Lazaro steering committee	1 Dec	Staff time; Xcel's battery rebate already funded	People on oxygen or powered devices losing power without warning

Xcel cut power to about 55,000 customers, mainly in Boulder County, on 6 April 2024. The city's wastewater plant nearly overflowed, and care homes were caught off guard. There have been four shutoffs since 2024. **Much of the medical protection already exists, but only for people enrolled in it.** Xcel gives Medical Certificate customers 72 hours' notice and offers a backup-battery rebate of up to $10,000 (up to 100% for income-qualified customers) in Tier 2 and 3 wildfire zones, from a reported budget of about $2M, first-come. The action has three parts. First, use HHS emPOWER counts for ZIPs 80301–80305 to find electricity-dependent residents who are not enrolled, and enrol them. Second, add at least two charging sites in south and east Boulder that can power oxygen and medical devices; on 14 March 2026 the only city site was North Boulder Rec, open 2–6pm. Third, run volunteer buddy lists for people with no car or on powered equipment in mobile-home parks and on one-road foothills streets. San Lazaro has about 1,000 residents and more than 140 children, speaks English, Spanish, and Bosnian, and has a steering committee that already organises. It is in unincorporated Boulder County, so this part is county-led.*

* Suggested formulation: check with a health worker before sharing.

Transferability: native. Xcel programme details come from news reports; the tariff and the PUC docket could not be read. Xcel will not share its customer list without consent, and emPOWER gives counts by ZIP, not names, so individual outreach needs an HHS data request. No death was reported from the 2024 shutoff.

5. Guarantee a bed on every Critical Weather night, and pause camping enforcement on those nights

PARTIALLY CONFIRMED | FRONTLINE · SKEPTIC · EMERGENCY MGR · PUBLIC HEALTH · PHILANTHROPIST | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
City Housing & Human Services and All Roads (4869 Broadway); Boulder Police and the city manager for enforcement	Before the first Critical Weather night	$40k conditional (see funding); enforcement pause costs nothing	Exposure deaths among people turned away or moved into hiding

The trigger is not the problem the lenses thought it was. The city's current page lists Critical Weather Nights as overnight shelter at All Roads when the forecast low is 20°F or below, or there are 6 inches or more of snow, or gusts above 70 mph. Arrival is before 7pm, and "beds are limited… no guarantee." Beds fell from 180 to 160 after a 30% funding cut in April 2025, and All Roads expected nightly turn-aways. Normal rules allow one night, then a Coordinated Entry assessment. Camping citations along the Boulder Creek path roughly doubled in early 2025, to at least 115 against about 60. Officers are told not to say where people can go. The action has two parts. On Critical Weather nights, waive Coordinated Entry and the one-night limit and open an overflow site so nobody is turned away. On the same nights, pause clearances and citations, and pair outreach with a guaranteed bed. In 2011–2018, 81 unhoused people died in the City of Boulder, and hypothermia was a factor in 18% of those deaths.

* Suggested formulation: check with a health worker before sharing.

Transferability: native. The current bed count was not confirmed, and the bed cut is 18 months old. Clearances at 2–3am rest on one resident's account. Pausing enforcement needs no ordinance and stays legal after *Grants Pass* (2024). Confirm whether the city page's "20°F during the day" wording is a typo. The November 2024 protocol used 10°F for day services and 0°F for the Extreme Weather Shelter.

6. Fund the county's existing falls-prevention classes, and hand out ice cleats through them

PARTIALLY CONFIRMED | PUBLIC HEALTH | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Boulder County Area Agency on Aging	Cleats out by 1 Dec	*Unfunded*, small (fee subsidies and added places)	Falls, the largest accident category in coroner cases

Falls were 74 of 330 Boulder County coroner cases in 2025, all classed as accidents. The Area Agency on Aging already runs a falls-prevention series: Tai Chi for Arthritis and to Prevent Falls ($35), N'Balance, Balance for Body and Mind, safe-falling classes, and Spanish-language sessions. Subsidise fees, add places, and hand out cleats through the classes. Exercise cuts the rate of falls by about 23% (Sherrington, Cochrane 2019). Cleats halved outdoor falls in one small Wisconsin trial (McKiernan 2005).

Transferability: moderate. The coroner report gives no age split and no ice field. Neither study was re-checked in this run (both were checked earlier the same day for Lake Geneva). No link between El Niño and falls is established.

7. After each heavy snowfall, warn men about shoveling and the heart

PARTIALLY CONFIRMED | PUBLIC HEALTH | INVENTORY: WINTER WEATHER ADVISORIES (ANAND 2025), INDIRECT

OWNER	BY WHEN	COST	PREVENTS
Boulder County Public Health via ODM alert channels	Message drafted by 1 Dec; sent the day after 20 cm or more	Staff time	Heart-attack deaths after heavy snow, the one El Niño-linked killer here

* Suggested formulation: check with a health worker before sharing.

This is the one action tied directly to El Niño's local signal, heavy wet snow. In Quebec (1981–2014), men had 1.34 times the odds of dying of a heart attack the day after 20 cm or more of snow (95% CI 1.26–1.42), and women showed no association (Auger et al., *CMAJ* 2017). On 13–15 March 2024, during the last strong El Niño, Boulder got 21.9 inches. Pre-draft a message for the day after any 20 cm (8-inch) snowfall, and point people to volunteer shoveling help once a programme is confirmed.

Transferability: the study is from sea-level Quebec. Altitude is untested, and the study did not single out older men. No Boulder volunteer shoveling programme was found.

8. Ask the coroner to restore the unhoused-death table and add an ice-or-snow flag before winter

PARTIALLY CONFIRMED | PUBLIC HEALTH · HISTORIAN | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Boulder County Coroner	Before 1 Dec	Staff time	Planning blind: no one can attribute a cold-season death today

The coroner's reports tracked unhoused deaths through about 2020, when they recorded 5 hypothermia deaths. The current dashboard lists only age, sex, and location, and the 2025 summary has no housing-status or ice field. Restore the published table, and flag ice or snow in accidental falls and exposure deaths. With about 3 exposure deaths a year, any programme's effect is otherwise unmeasurable.

Transferability: native. It saves no life directly, so it ranks low, but every other action here is unevaluable without it.

9. Household money: LEAP application drives from 1 November, furnace repair, and the replacement-cost disclosures the law already requires

PARTIALLY CONFIRMED | ECONOMIST | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Colorado CDHS (LEAP); Energy Outreach Colorado; Colorado Division of Insurance	Documents ready in Oct; drives from 1 Nov	Benefits already funded	Heat loss in cold homes, and the next fire's underinsurance gap

* Suggested formulation: check with a health worker before sharing.

LEAP's season runs 1 November 2026 to 30 April 2027, and applications open only on 1 November, so “pre-enrol before November” is wrong. Prepare documents now and run drives from 1 November. Energy Outreach Colorado served 40,789 households in 2025 and offers emergency furnace repair. On insurance, HB23-1174 already requires insurers to disclose replacement-cost information before they issue or renew a policy, and to offer extended replacement cost of at least 50%. The action is to publicise it, not to ask for something new. After the Marshall Fire, FEMA paid individual households $2.13M across 948 approvals, against $45.1M in public assistance and $11.3M in hazard mitigation.

Transferability: native. The Division of Insurance's Marshall underinsurance analysis (36–67% underinsured) could not be read and is not used. Colorado PUC winter disconnection protections were not verified. This protects household finances more than lives.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Office of Disaster Mgmt	BRETSA (911)	Boulder Fire-Rescue	County Public Health (The Works)	City Housing & Human Services	All Roads	Boulder Police	Xcel Energy	County Area Agency on Aging	County Coroner	San Lazaro steering committee	Energy Outreach & CDHS (LEAP)	Colorado Div. of Insurance	Community Foundation Boulder County
1. Bystander naloxone and training			C	A/R	C	R	I			C				C
2. Pre-approved alerts, WEA drill, surge senders	A	R	C	I			C	C			I			
3. Wind days are fire days	R	I	A/R				I	C			I			
4. Shutoff medical gap, charging sites, buddy lists	R		A	R	C			R	C		R			C
5. Bed guarantee and enforcement pause	C			C	A	R	R			I				C
6. Falls classes and cleats			C	C					A/R					
7. Snow-day heart warning	R		C	A/R					C					
8. Coroner housing and ice fields	I			C	C					A/R				
9. LEAP drives, furnace repair, insurance disclosure					R			I	C		R	A/R	R	

What the chart shows. **ODM and Fire-Rescue carry the wind rows (2–4)**, and those rows hold the plan's largest documented failure, the Marshall alert gap. **Public Health is accountable for the two rows that address the largest death counts** (naloxone and the snow-day warning). **The San Lazaro steering committee is the only resident body with an R**. It already organises in three languages, but it is in unincorporated county, outside every city row. **Row 5 sits with the city and the police together**: a bed guarantee without an enforcement pause, or the reverse, does not work.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. Most of Boulder's protective actions cost staff time; the money below is small on purpose.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$50k	$90k	~$200k	All Roads' current bed count and 2027 county award, and whether the

* Suggested formulation: check with a health worker before sharing.

state naloxone fund is rationing The Works

Floor: $25k for naloxone distribution plus $25k for a trigger fund, which totals $50k. **Target:** $25k + $40k for All Roads + $25k, which totals $90k. **Ceiling:** $50k + $75k + $75k, which totals $200k. The target fell 22% from the lens's $115k. The naloxone line moved from $50k of kits to $25k of distribution, because the state fund supplies kits free.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Boulder County Public Health, The Works	Action 1: bystander naloxone distribution and training in the city; shelter stock	$25k (was $50k for kits)	Restricted grant	Kits free from the CDPHE Naloxone Bulk Purchase Fund; Coffin & Sullivan 1 death averted per 227 kits (US model, not re-checked)	PARTIALLY CONFIRMED
All Roads (formerly Boulder Shelter for the Homeless)	Action 5: bed guarantee and overflow on Critical Weather nights	$40k, conditional	Unrestricted winter gift	Beds cut 180 → 160 in April 2025 after a 30% cut. Release only after the current bed count and 2027 county award (from an ~$8.5M Community Partnership Grant round) are confirmed	PARTIALLY CONFIRMED
Community Foundation Boulder County	Action 4: pre-positioned trigger fund for warming and power for medically vulnerable people in windstorms and shutoffs	$25k	Donor fund released on a shutoff or High Wind Warning	No cost-per-death study exists. CFBC's current Resilience Fund is for nonprofit collaboration, so this fund would be new	PARTIALLY CONFIRMED

The funder's one thing: El Niño is the wrong reason to give in Boulder. In 2025 drugs killed about 25 times as many people here as environmental exposure did, so the marginal dollar belongs in bystander naloxone in the city, where 13 of the first 20 fentanyl deaths of 2026 occurred. Weather money buys visibility more than lives.

* Suggested formulation: check with a health worker before sharing.

DO NOT DO THIS CONTRAINDICATED

Advice that circulates for this hazard and would cost lives or waste the window here.

Stand down fire readiness because “El Niño means a snowy winter”

AVOID (SKEPTIC, HISTORIAN; VERIFIED IN ACTION 3)

El Niño raises Boulder's odds of wetter months, mostly through late-winter upslope snow. It does not touch the November–February downslope windstorms, which carry 74% of 70-mph-plus events. The Marshall Fire burned on 30 December on a snowless grass fuel bed. A snowy forecast is how a dry, windy week between storms gets missed.

Instead: trigger on Red Flag and High Wind Warnings all winter (action 3).

Buy naloxone kits

CONTRAINDICATED AS A PURCHASE

The Philanthropist's top placement was $50k for about 1,100 kits. Colorado's Naloxone Bulk Purchase Fund already supplies eligible organisations free.

Instead: fund the people who hand kits out and train bystanders (action 1), and buy kits only to fill a gap The Works documents.

Clear camps on the coldest nights, or run outreach alongside enforcement

AVOID (FRONTLINE, SKEPTIC; PARTIALLY VERIFIED)

On Critical Weather nights the shelter has limited beds and no guarantee, so moving people without a bed raises their exposure. The Frontline lens's warning holds: the pink tags taught people to hide, and the person who most needs a warm bed on the coldest night is the one who moved off the path so the city could not find them.

Instead: pause enforcement on Critical Weather nights, and keep outreach lists away from enforcement (action 5).

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
May–June 1894	Boulder Creek flood, slow-rising	1 dead	n/a
Sept 1938	Flood, including South Boulder Creek	Not found	Flood zone known; fix not built for 88 years
23 Jan 1971 / Jan 1982	Downslope windstorms: 147 mph and 137 mph at NCAR	1982: 40% of buildings damaged (lens)	n/a
Dec 1982	El Niño: 23.5 in of snow in the month	Not found	n/a
Oct 1997	El Niño: 30.1 in; 1997 the snowiest year on record (158.7 in)	Not found	n/a
Mar 2003	El Niño: 34.7 in in the month; FEMA snow declaration	Not found	n/a
2011–2018	Unhoused deaths, City of Boulder	81; hypothermia a factor in 18%	Coroner tracking continued to ~2020, then dropped
Sept 2013	Flood: 9.08 in in one day (neutral ENSO)	3 dead in Boulder County	South Boulder Creek fix began 31 Aug 2026; $76.5M, 3–5 years
30 Dec 2021	Marshall Fire (La Niña): grass, 115-mph gusts, power line	2 dead; 1,084 homes	WEA now listed as automatic; HB23-1174 insurance disclosure
13–15 Mar 2024	Strong El Niño: 21.9 in, heavy and wet	Not found	n/a
6 Apr 2024	Xcel shutoff, ~55,000 customers	No deaths reported; wastewater plant near overflow	PUC investigation; undergrounding began 2026

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Apr 2025	All Roads beds cut 180 → 160; camping citations roughly doubled	n/a	n/a
2025	Coroner year	77 drug/alcohol, 3 exposure, 74 falls (of 330 cases)	n/a
Dec 2025 – Mar 2026	Shutoff 20 Dec; Hwy 287/Dillon Rd fire evacuations 5 Jan; shutoff 14 Mar	Not found	One in-city charging site

The recurrence nobody has noticed: Boulder's El Niño winters bring its biggest snow months, and its disasters arrive out of that season: the September floods and the late-December fire in a La Niña year. The region's memory is reliable and its building is slow. It knew where South Boulder Creek would flood from 1938, and broke ground in 2026.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Spend on weather** (Emergency manager, Skeptic, Frontline) versus **spend on overdose** (Public health, Philanthropist). The coroner's numbers favour overdose on volume (77 against 3). But the wind-fire tail risk is catastrophic, and its fixes are mostly staff time. This report ranks naloxone first and wind actions next, because they compete for different resources.
- **WEA is still not in use** (Skeptic) versus **WEA is now listed as automatic** (verifier). The policy is confirmed; use in a real fire since 2022 is not.
- **Lower the shelter trigger** (Frontline, Skeptic) versus **the trigger is already 20°F or below, and the gap is beds** (verifier). Settled in favour of capacity, pending confirmation of the city page's wording.

WHAT NOBODY COVERED

BLIND SPOT

Only the Historian raised **South Boulder Creek**, where about 2,300 people and 260 structures stay exposed through a 3–5 year construction period that began on 31 August 2026. It is a spring and summer flood risk, outside this window, and it was not verified. No lens covered **CU Boulder students** off campus in shutoffs and snow, or **avalanche and backcountry deaths** in the foothills, which a snowy El Niño season can raise. No lens found **carbon monoxide** data for the county, though shutoffs and heavy-snow outages are when generators come out.

* Suggested formulation: check with a health worker before sharing.

HOW TO READ THIS

METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people, and are not neighbourhood figures. Boulder County holds about 330,000. The El Niño projection covers *heat only*; fire, wind, snow, and cold are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Boulder, Colorado, read as the City of Boulder, a city of about 105,000. That is coarser than the neighbourhood scale this method is built for. Actions are aimed at named sub-areas (the Boulder Creek path, the west-edge wildland–urban interface, mobile-home parks) where the record allowed. Climate Impact Lab impact region, as this report records it: Boulder County, CO, *inferred* rather than resolved on the Impact Map. Where lenses used administrative data, the unit is named: Boulder County (coroner), the City of Boulder (unhoused deaths 2011–18, city fentanyl deaths 2026), and ZIPs 80301–80305 (emPOWER). San Lazaro is outside city limits.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias before it counted. Here the convergence that El Niño is the wrong frame rests on the county's own coroner data and NOAA's wind record, not on shared literature, so it held. The convergence on a lower shelter trigger did not hold, because the lenses read the November 2024 protocol, not the current one. Contradictions were kept. Actions were ranked by expected lives saved per unit of effort, each given a payer or marked unfunded, and the funding placements reconciled against them.

* Suggested formulation: check with a health worker before sharing.

- **Verification.** Five separate verifier agents checked 14 recommendations and the three funding placements for evidence, transferability, and feasibility. Financial actions were also checked for availability, payer, trigger, and existing implementation, and the envelope for real recipients, unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not re-opened in this run. The inventory's US filter (19 entries) was read in a browser earlier the same day for the Gaslamp Quarter briefing, and those titles were used here. *Winter weather advisories* (Anand 2025) bears indirectly on action 7, *Change in forecast quality* (Shrader et al. 2023) on action 2, and *Substance abuse regulation policies* loosely on action 1. No US entry covers alerting systems, power shutoffs, or shelter capacity.
- **Health wording.** Passages where a wording error could hurt someone (naloxone, and oxygen and medical equipment in an outage) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total without the division being shown and labelled. The Economist's $34M underinsurance figure (344 × $98,967) was such a derivation, from a source that could not be read, and is excluded. No local figure is invented; where data is thin, the report says so.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. **The shared web-search budget was exhausted before this run began**, so every lens and verifier fetched pages directly, and xcelenergy.com, puc.colorado.gov, doi.colorado.gov, and several city pages could not be read. The Marshall after-action report itself was not read; its findings are cited through press coverage. The 2026–27 shelter protocol and the current All Roads bed count were not confirmed.

References · verified by Phase 5 unless marked. Boulder County Coroner 2025 summary (verified) · Coroner 2025 drug report (verified) · Coroner 2026 biannual drug report (verified) · CDPHE Naloxone Bulk Purchase Fund (verified) · The Works Program (verified) · Opioid settlement grants (corrected) · Marshall alerts, Denver7 (verified) · Marshall alert gaps, Colorado Politics (verified) · Marshall after-action, CBS Colorado (verified) · ODM emergency alerts (WEA automatic) (verified) · ODM emergency status (verified) · 14 Mar 2026 shutoff sites (verified) · NOAA Boulder wind record (corrected) · NOAA Boulder snow record (lens, not re-verified) · NWS Boulder El Niño outlook (verified) · CPC ENSO advisory, 8 Oct 2026 (verified) · Marshall Fire (verified) · Xcel medical notice and battery rebate, Denver7 (verified, news) · Shutoffs and undergrounding, BRL (verified) · April 2024 shutoff, Colorado Sun (verified) · HHS emPOWER (verified) · San Lazaro, BRL (verified) · City homelessness services (Critical Weather Nights) (verified) · Nov 2024 severe-weather protocol (verified) · All Roads (verified) · All Roads bed cut, Longmont Leader

* Suggested formulation: check with a health worker before sharing.

(verified) · Camping enforcement, BRL (verified) · Unhoused deaths 2011–18, Boulder Beat (corrected) · Coroner 2020 unhoused deaths, Boulder Beat (verified) · Area Agency on Aging falls prevention (verified) · Auger et al., CMAJ 2017 (verified) · LEAP (verified) · Energy Outreach Colorado (verified) · HB23-1174 (verified) · FEMA DR-4634 summary (verified) · County Community Partnership Grants (verified) · Community Foundation Boulder County (verified) · South Boulder Creek flood mitigation (lens, not re-verified) · 2013 flood, NOAA (lens, not re-verified)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (US)
Generated 8 October 2026 for Boulder, Colorado. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.